AnimeGAN2 Video Processing Pipeline

COMPLETE SETUP, EXECUTION & ASSEMBLY GUIDE

レーナー

Environment	Python Version	Hardware / Acceleration	Target Model
WSL / Ubuntu 24.04	Python 3.10 (Conda `animeenv`)	NVIDIA GPU (CUDA 12.1)	AnimeGAN2 (Paprika Style)

1. Environment Creation & ToS Acceptance

Ubuntu 24.04 ships with Python 3.14, which lacks PyTorch wheel support. Conda is used to establish a Python 3.10 virtual environment.

```
# Accept Conda Terms of Service if prompted /home/admin1/miniconda/bin/conda tos accept --
override-channels --channel https://repo.anaconda.com/pkgs/main
/home/admin1/miniconda/bin/conda tos accept --override-channels --channel
https://repo.anaconda.com/pkgs/r # Create the Python 3.10 environment
/home/admin1/miniconda/bin/conda create -n animeenv python=3.10 -y
```

2. CUDA PyTorch & Package Installation

PyTorch with CUDA 12.1 support must be installed from official PyTorch wheels, followed by pipeline dependencies.

```
# Install PyTorch with CUDA 12.1 support /home/admin1/miniconda/envs/animeenv/bin/pip
install torch==2.5.1 torchvision==0.20.1 --index-url https://download.pytorch.org/whl/cu121
# Install requirements and essential utilities /home/admin1/miniconda/envs/animeenv/bin/pip
install -r /mnt/d/aipipeline/requirements.txt /home/admin1/miniconda/envs/animeenv/bin/pip
install tqdm Pillow # Install system FFmpeg (required for video/audio processing) sudo apt
update && sudo apt install -y ffmpeg
```

3. Pipeline & Stage 4 Stylization Execution

Execute the core AnimeGAN2 frame processing pipeline using GPU acceleration.

```
cd /mnt/d/aipipeline/animegan2-pytorch /home/admin1/miniconda/envs/animeenv/bin/python3
convert_video_anime.py
```

Stage 4 Output: Processed frames are stored in 4-digit zero-padded JPEG format (`0001.jpg` to `80972.jpg`) under `/mnt/d/aipipeline/pipeline_output/stage5_anime_frames/`.

4. Final Video & Audio Muxing (FFmpeg)

Combine the stylized frames with the original dubbed audio track into a single MP4 file.

```
cd /mnt/d/aipipeline ffmpeg -y -framerate 30 -i pipeline_output/stage5_anime_frames/%04d.jpg
\ -i stage1/final_dubbed_video.mp4 \ -c:v libx264 -pix_fmt yuv420p -c:a aac -shortest \
pipeline_output/final_anime_video.mp4
```

5. Automated Recovery & Assembly Scripts

Two shell scripts were created to automate environment recovery and final video assembly.

```
# Execute full automated environment recovery & launch /mnt/d/aipipeline/recover_and_run.sh
# Execute standalone video/audio muxing script /mnt/d/aipipeline/mux_video.sh
```